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8	Using the swing components, design the frame for shopping a book that accepts book code, book name, and Price. Calculate the discount on code as follows. <table><tr><td>Code</td><td>Discount rate</td></tr><tr><td>101</td><td>15%</td></tr><tr><td>102</td><td>20%</td></tr><tr><td>103</td><td>25%</td></tr><tr><td>Any other</td><td>5%</td></tr></table> Find the discount amount and Net bill amount. Display the bill	Code	Discount rate	101	15%	102	20%	103	25%	Any other	5%	
Code	Discount rate											
101	15%											
102	20%											
103	25%											
Any other	5%											

PART - A

1. Program to accept student name and marks in three subjects. Find the total marks, average and grade (depending on the average marks).

Code:

```
import java.io.*;

public class Grade {

    public static void main(String[] args) throws IOException {

        String name, g;

        float m1, m2, m3, total, average;

        DataInputStream dis = new DataInputStream(System.in);

        System.out.print("Enter student name: ");

        name = dis.readLine();

        System.out.println("Enter marks in 3 subjects: ");

        m1 = Float.parseFloat(dis.readLine());

        m2 = Float.parseFloat(dis.readLine());

        m3 = Float.parseFloat(dis.readLine());

        System.out.println("\nStudent name: " + name);

        System.out.println("Marks in subject 1: " + m1);

        System.out.println("Marks in subject 2: " + m2);

        System.out.println("Marks in subject 3: " + m3);

        total = m1 + m2 + m3;

        System.out.println("Total = " + total);

        average = total / 3;

        System.out.println("Average = " + average);

        if (m1 <= 35 || m2 <= 35 || m3 <= 35)

            g = "Fail";

        else if (average >= 85)
```

```
        g = "A";
    else if (average >= 70)
        g = "B";
    else if (average >= 60)
        g = "C";
    else
        g = "D";
    System.out.println("Grade = " + g);
}
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac Grade.java
Note: Grade.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.

D:\Documents\Java\LAB PROGRAMS>java Grade
Enter student name: Ashwith Frank
Enter marks in 3 subjects:
94
89
97

Student name: Ashwith Frank
Marks in subject 1: 94.0
Marks in subject 2: 89.0
Marks in subject 3: 97.0
Total = 280.0
Average = 93.333336
Grade = A
```

2. Program, which reads two numbers having same number of digits. The program outputs the sum of product of corresponding digits.(Hint Input 327 and 539 output $3 \times 5 + 2 \times 3 + 7 \times 9 = 84$)

Code:

```
import java.io.*;

public class SumOfProduct {

    public static void main(String[] args) throws IOException {

        int result = 0;

        DataInputStream dis = new DataInputStream(System.in);

        System.out.println("Enter two numbers of same digits:");

        int num1 = Integer.parseInt(dis.readLine());

        int num2 = Integer.parseInt(dis.readLine());

        while (num1 > 0 && num2 > 0) {

            result = result + ((num1 % 10) * (num2 % 10));

            num1 /= 10;

            num2 /= 10;

        }

        System.out.println("Sum of product of corresponding digits: " + result);

    }

}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac SumOfProduct.java
Note: SumOfProduct.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.

D:\Documents\Java\LAB PROGRAMS>java SumOfProduct
Enter two numbers of same digits:
327
539
Sum of product of corresponding digits: 84
```

3. Program to input Start and End limits and print all Fibonacci numbers between the ranges. (Use for loop)

Code:

```
import java.util.Scanner;

class Fibonacci {

    public static void main(String args[]) {

        int l, u, f1 = 0, f2 = 1, f3;

        Scanner s = new Scanner(System.in);

        System.out.println("Enter the lower and upper limit:");

        l = s.nextInt();

        u = s.nextInt();

        System.out.println("Fibonacci numbers in the given range: ");

        for (int i = f1; i <= u; i++) {

            if (f1 >= 1 && f1 <= u)

                System.out.print(f1 + " ");

            f3 = f1 + f2;

            f1 = f2;

            f2 = f3;

        }

    }

}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac Fibonacci.java

D:\Documents\Java\LAB PROGRAMS>java Fibonacci
Enter the lower and upper limit:
0 20
Fibonacci numbers in the given range:
0 1 1 2 3 5 8 13
D:\Documents\Java\LAB PROGRAMS>
```


4. Define a class named Pay with data members String name, double salary, double da, double hra, double pf, double grossSal, double netSal and methods: Pay(String n, double s) - Parameterized constructor to initialize the data members, void calculate() - to calculate the following salary components, and void display() - to display the employee name, salary and all salary components.

- **Dearness Allowance = 15% of salary**
- **House Rent Allowance = 10% of salary**
- **Provident Fund = 12% of salary**
- **Gross Salary = Salary + Dearness Allowance + House Rent Allowance**
- **Net Salary = Gross Salary - Provident Fund**

Write a main method to create object of the class and call the methods to compute and display the salary details. [class basics]

Code:

```
import java.io.*;

public class Pay {

    String name;

    double salary;

    double da;

    double hra;

    double pf;

    double grossSal;

    double netSal;

    public Pay(String n, double s) {

        name = n;

        salary = s;

        da = 0;

        hra = 0;
```

```
    pf = 0;
    grossSal = 0;
    netSal = 0;
}

void calculate() {
    da = salary * 15.0 / 100;
    hra = salary * 10.0 / 100;
    pf = salary * 12.0 / 100;
    grossSal = salary + da + hra;
    netSal = grossSal - pf;
}

void display() {
    System.out.println("\nEmployee Name : " + name);
    System.out.println("Salary : " + salary);
    System.out.println("Dearness Allowance : " + da);
    System.out.println("House Rent Allowance : " + hra);
    System.out.println("Provident Fund: " + pf);
    System.out.println("Gross Salary : " + grossSal);
    System.out.println("Net Salary : " + netSal);
}

public static void main(String args[]) throws IOException {
    DataInputStream dis = new DataInputStream(System.in);
    System.out.print("Enter Employee Name: ");
```

```
String empName = dis.readLine();

System.out.print("Enter Salary: ");

double empSal = Double.parseDouble(dis.readLine());

Pay obj = new Pay(empName, empSal);

obj.calculate();

obj.display();

}

}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac Pay.java
Note: Pay.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.

D:\Documents\Java\LAB PROGRAMS>java Pay
Enter Employee Name: Ashwith Frank
Enter Salary: 25000

Employee Name : Ashwith Frank
Salary : 25000.0
Dearness Allowance : 3750.0
House Rent Allowance : 2500.0
Provident Fund: 3000.0
Gross Salary : 31250.0
Net Salary : 28250.0
```

5. Program to create a class DISTANCE with the data members feet and inches. Use a constructor to read the data and a member function Sum () to add two distances by using objects as method arguments and show the result. (Input and output of inches should be less than 12.).

Code:

```
import java.util.*;

class Distance {

    int feet;

    int inches;

    public Distance() {

        feet=0;

        inches=0;

    }

    public void getDistance() {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter feet: ");

        feet = sc.nextInt();

        System.out.print("Enter inches: ");

        inches = sc.nextInt();

        System.out.println();

    }

    public void showDistance() {

        System.out.println("Feet: " + feet + "\nInches: " + inches);

    }

}
```

```
public void Sum(Distance D1, Distance D2) {  
    inches = D1.inches + D2.inches;  
    feet = D1.feet + D2.feet;  
}  
}
```

```
public class AddTwoDistances {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        Distance D1 = new Distance();  
        Distance D2 = new Distance();  
        Distance D3 = new Distance();  
        System.out.println("Enter first distance: ");  
        D1.getDistance();  
        System.out.println("Enter second distance: ");  
        D2.getDistance();  
        D3.Sum(D1, D2);  
        if (D3.inches > 12) {  
            D3.feet++;  
            D3.inches = D3.inches - 12;  
        }  
        System.out.println("Total distance is: ");  
        D3.showDistance();  
    }  
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac AddTwoDistances.java

D:\Documents\Java\LAB PROGRAMS>java AddTwoDistances
Enter first distance:
Enter feet: 15
Enter inches: 4

Enter second distance:
Enter feet: 21
Enter inches: 5

Total distance is:
Feet: 36
Inches: 9
```

6. Program to create a class “Matrix” that would contain integer values having varied numbers of columns for each row. Print row-wise sum.

Code:

```
import java.util.Scanner;

public class Matrix {

    int m, n;

    int a[][];

    Matrix(short rows) {

        m = rows;

        n = 0;

        a = new int[m][];

    }

    void read() {

        int i, j;

        Scanner sc = new Scanner(System.in);

        for (i = 0; i < m; i++) {

            System.out.println("Enter the number of coloumns in row " + (i + 1));

            short n = sc.nextShort();

            a[i] = new int[n];

        }

        for (i = 0; i < m; i++) {

            System.out.println("Enter the elements of row " + (i + 1));

            for (j = 0; j < a[i].length; j++) {
```

```
        a[i][j] = sc.nextInt();
    }
}
}
```

```
void display() {
    int i, j;
    System.out.println("\nThe given Matrix");
    for (i = 0; i < m; i++) {
        for (j = 0; j < a[i].length; j++)
            System.out.print(a[i][j] + " ");
        System.out.println();
    }
    System.out.println();
    for (i = 0; i < m; i++) {
        int sum = 0;
        for (j = 0; j < a[i].length; j++)
            sum = sum + a[i][j];
        System.out.println("Sum of row " + (i + 1) + " = " + sum);
    }
}
```

```
public static void main(String args[]) {
    short m;

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter the number of rows: ");
```



```
m = sc.nextShort();  
  
Matrix obj = new Matrix(m);  
  
obj.read();  
  
obj.display();  
  
}  
  
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]  
(c) Microsoft Corporation. All rights reserved.  
  
D:\Documents\Java\LAB PROGRAMS>javac Matrix.java  
  
D:\Documents\Java\LAB PROGRAMS>java Matrix  
Enter the number of rows: 3  
Enter the number of coloumns in row 1  
2  
Enter the number of coloumns in row 2  
3  
Enter the number of coloumns in row 3  
4  
Enter the elements of row 1  
1 2  
Enter the elements of row 2  
3 4 5  
Enter the elements of row 3  
6 7 8 9  
  
The given Matrix  
1 2  
3 4 5  
6 7 8 9  
  
Sum of row 1 = 3  
Sum of row 2 = 12  
Sum of row 3 = 30
```

7. Program to extract portion of character string and print extracted string. Assume that 'n' characters extracted starting from mth character position.

Code:

```
import java.util.*;

class ExtractString {

    public static void main(String args[]) {

        Scanner sc = new Scanner(System.in);

        String str1, str2;

        int m = 0, n = 0;

        System.out.print("Enter a String: ");

        str1 = sc.nextLine();

        System.out.print("Enter numbers of characters to be extracted from string:");

        m = sc.nextInt();

        System.out.print("Enter Starting Index: ");

        n = sc.nextInt();

        System.out.println("The given string is: " + str1);

        str2 = str1.substring(n, (m + n));

        System.out.println("The extracted from string is: " + str2);

    }

}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac ExtractString.java

D:\Documents\Java\LAB PROGRAMS>java ExtractString
Enter a String: Ashwith Frank
Enter numbers of characters to be extracted from string: 7
Enter Starting Index: 0
The given string is: Ashwith Frank
The extracted from string is: Ashwith
```

8. Program to add, remove and display elements of a Vector.

Code:

```
import java.util.Vector;
import java.util.Scanner;
public class VectorList {
    public static void main(String args[]) {
        Vector v = new Vector(6, 2);
        String item;
        int ch = 0, i = 0, n;
        Scanner sc = new Scanner(System.in);
        while (ch != 5) {
            System.out.println("\n1.Append \n2.Insert \n3.Delete \n4.Display \n5.Exit");
            System.out.print("Enter your choice: ");
            ch = sc.nextInt();
            switch (ch) {
                case 1:
                    System.out.print("Enter the item to append: ");
                    item = sc.next();
                    v.add(item);
                    break;
                case 2:
                    System.out.print("Enter the item to insert: ");
                    item = sc.next();
                    System.out.print("Enter the index: ");
                    i = sc.nextInt();
```

```

        v.insertElementAt(item, i);
        break;
case 3:
    System.out.print("Enter the item to delete: ");
    item = sc.next();
    if (v.remove(item) == true)
        System.out.println(item + " deleted from the list");
    else
        System.out.println(item + " not found in the list");
    break;
case 4:
    n = v.size();
    for (i = 0; i < n; i++) {
        item = (String) v.elementAt(i);
        System.out.println(item);
    }
}
}
}
}

```

Output:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

D:\Documents\Java\LAB PROGRAMS>javac VectorList.java
Note: VectorList.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.

D:\Documents\Java\LAB PROGRAMS>java VectorList

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 1
Enter the item to append: Apple

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 1
Enter the item to append: Banana

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 2
Enter the item to insert: Mango
Enter the index: 1

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 3
Enter the item to delete: Apple
Apple deleted from the list

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 4
Mango
Banana

1.Append
2.Insert
3.Delete
4.Display
5.Exit
Enter your choice: 5
```

PART – B

1. Create a class named 'Member' having data members: Name, Age, Phone Number, Place and Salary. It also has a method named 'printSalary' which prints the salary of the members. Two classes 'Employee' and 'Manager' inherit the 'Member' class. The 'Employee' and 'Manager' classes have data members 'specialization' and 'department' respectively. Now, assign name, age, phone number, address and salary to an employee and a manager by making an object of both classes and print the same.

Code:

```
import java.util.Scanner;

class Member {

    String name;

    int age;

    String phone_no;

    String place;

    long salary;

    public void readMemData() {

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter employee name, age, phone number, place and salary");

        name = sc.next();

        age = sc.nextInt();

        phone_no = sc.next();

        place = sc.next();

        salary = sc.nextLong();

    }

    public void printMemData() {

        System.out.println("Name : " + name);

        System.out.println("Age : " + age);

        System.out.println("Phone No : " + phone_no);

        System.out.println("Place : " + place);

    }

    public void printSalary() {

        System.out.println("Salary : " + salary);

    }

}
```

```

    }
}
class Employee extends Member {
    String specialization;
    void readEmpSp() {
        readMemData();
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter specialization");
        specialization = sc.next();
    }
    void printEmpSp() {
        printMemData();
        System.out.println("Specialization : " + specialization);
    }
}
class Manager extends Member {
    String dept;
    void readManDept() {
        readMemData();
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the department");
        dept = sc.next();
    }
    void printManDept() {
        printMemData();
        System.out.println("Department : " + dept);
    }
}
class SingleDemo {

```



```
public static void main(String args[]) {  
    Employee e1 = new Employee();  
    Manager m1 = new Manager();  
    System.out.println("*****Employee Details*****");  
    System.out.println();  
    e1.readEmpSp();  
    System.out.println("-----");  
    e1.printEmpSp();  
    e1.printSalary();  
    System.out.println(".....");  
    System.out.println("*****Manager Details*****");  
    System.out.println();  
    m1.readManDept();  
    System.out.println("-----");  
    m1.printManDept();  
    m1.printSalary();  
    System.out.println(".....");  
}  
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

G:\Ashwith\Java\LAB PROGRAMS\PART B\B1. Member>javac SingleDemo.java

G:\Ashwith\Java\LAB PROGRAMS\PART B\B1. Member>java SingleDemo
*****Employee Details*****

Enter employee name, age, phone number, place and salary
Rakesh
25
9876543210
Madanthyar
25000
Enter specialization
Developer
-----
Name : Rakesh
Age : 25
Phone No : 9876543210
Place : Madanthyar
Specialization : Developer
Salary : 25000
.....
*****Manager Details*****

Enter employee name, age, phone number, place and salary
Kumar
30
9753186420
Mangalore
40000
Enter the department

Finance
-----
Name : Kumar
Age : 30
Phone No : 9753186420
Place : Mangalore
Department : Finance
Salary : 40000
.....
```

2. Program to implement the following class hierarchy:

Student: id, name

Student Exam (derived from Student): Marks of 3 subjects, total marks

Student Result (derived from Student Exam): percentage, grade

Define appropriate methods to accept and calculate grade based on existing criteria and display details of N students

Coding:

```
import java.util.Scanner;

class Student {

    Scanner sc = new Scanner(System.in);

    int id;

    String name;

    void getdata() {

        System.out.println("Enter Student ID and Name");

        id = sc.nextInt();

        name = sc.next();

    }

    void putdata() {

        System.out.println("Student ID : " + id);

        System.out.println("Name : " + name);

    }

}

class StudentExam extends Student {

    Scanner sc = new Scanner(System.in);

    float mark1, mark2, mark3, total;

    void getMarks() {

        getdata();

        System.out.println("Enter marks in 3 subjects");

        mark1 = sc.nextInt();
```

```

        mark2 = sc.nextInt();
        mark3 = sc.nextInt();
    }
    void putMarks() {
        putdata();
        System.out.println("*****");
        System.out.println("Marks obtained:");
        System.out.println("Subject 1 = " + mark1);
        System.out.println("subject 2 = " + mark2);
        System.out.println("subject 3 = " + mark3);
    }
    void Total() {
        total = mark1 + mark2 + mark3;
    }
}

class StudentResult extends StudentExam {
    void calculate() {
        Total();
        float percentage;
        String grade;
        percentage = total / 3;
        System.out.println("Percentage = " + percentage);
        if (mark1 <= 35 || mark2 <= 35 || mark3 <= 35)
            grade = "Fail";
        else if (percentage >= 85)
            grade = "A";
        else if (percentage >= 70)
            grade = "B";
        else if (percentage >= 60)

```

```
        grade = "C";
    else
        grade = "D";
    System.out.println("Grade = " + grade);
}
}
class Stud {
    public static void main(String args[]) {
        int cnt;
        Scanner sc = new Scanner(System.in);
        StudentResult ob = new StudentResult();
        System.out.println("Enter No. of Students You Want? : ");
        cnt = sc.nextInt();
        for (int i = 0; i < cnt; i++) {
            ob.getMarks();
            ob.putMarks();
            ob.calculate();
        }
    }
}
```

Output:

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

G:\Ashwith\Java\LAB PROGRAMS\PART B\B2. Student>javac Stud.java

G:\Ashwith\Java\LAB PROGRAMS\PART B\B2. Student>java Stud
Enter No. of Students You Want? :
2
Enter Student ID and Name
1001
Akash
Enter marks in 3 subjects
87 99 98
Student ID : 1001
Name : Akash
*****
Marks obtained:
Subject 1 = 87.0
subject 2 = 99.0
subject 3 = 98.0
Percentage = 94.666664
Grade = A
Enter Student ID and Name
1002
Ashwith
Enter marks in 3 subjects
23 45 52
Student ID : 1002
Name : Ashwith
*****
Marks obtained:
Subject 1 = 23.0
subject 2 = 45.0
subject 3 = 52.0
Percentage = 40.0
Grade = Fail
```

3. Write a program to calculate marks of a student using multiple inheritance implemented through interface. Class Student with data members rollNo, name, String cls and methods to set and put data. Create another class test extended by class Student with data members mark1, mark2, mark3 and methods to set and put data. Create interface sports with members sportsWt = 5 and putWt(). Now let the class results extends class test and implements interface sports. Write a Java program to read required data and display details in a neat format.

Coding:

```
import java.util.Scanner;

class student {

    int rollNo;

    String name;

    String cls;

    void getData(int n, String nm, String c) {

        rollNo = n;

        name = nm;

        cls = c;

    }

    void putData() {

        System.out.println("Roll No : " + rollNo);

        System.out.println("Name : " + name);

        System.out.println("Class : " + cls);

    }

}

class test extends student {

    int mark1, mark2, mark3;

    void getMarks(int m1, int m2, int m3) {

        mark1 = m1;

        mark2 = m2;

        mark3 = m3;
```

```

    }

    void putMarks() {
        System.out.println("*****");
        System.out.println("Marks obtained:");
        System.out.println("Subject 1 = " + mark1);
        System.out.println("subject 2 = " + mark2);
        System.out.println("subject 3 = " + mark3);
    }
}

interface sports {
    int sportsWt = 5;

    void putWt();
}

class results extends test implements sports {
    int total;

    public void putWt() {
        System.out.println("Sports Weight = " + sportsWt);
    }

    void display() {
        total = mark1 + mark2 + mark3 + sportsWt;
        putData();
        putMarks();
        putWt();

        System.out.println("*****");
        System.out.println("Total Score = " + total);
    }
}

class StudentInt {

```



```

public static void main(String args[]) {
    results r1 = new results();
    int m1, m2, m3, rno;
    String nm, cl;
    Scanner sc = new Scanner(System.in);
    System.out.println("Enter Roll No, Name and Class of the student ");
    rno = sc.nextInt();
    nm = sc.next();
    cl = sc.next();
    System.out.println("Enter Marks in three subjects ");
    m1 = sc.nextInt();
    m2 = sc.nextInt();
    m3 = sc.nextInt();
    System.out.println("*****");
    r1.getData(rno, nm, cl);
    r1.getMarks(m1, m2, m3);
    r1.display();
}
}

```

Output:

```

Microsoft Windows [Version 10.0.26200.8246]
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G:\Ashwith\Java\LAB PROGRAMS\PART B\B3. Student 2>javac StudentInt.java

G:\Ashwith\Java\LAB PROGRAMS\PART B\B3. Student 2>java StudentInt
Enter Roll No, Name and Class of the student
1001
Ashwith
IBCA
Enter Marks in three subjects
87 98 99
*****
Roll No : 1001
Name : Ashwith
Class : IBCA
*****
Marks obtained:
Subject 1 = 87
subject 2 = 98
subject 3 = 99
Sports Weight = 5
*****
Total Score = 289

```

4. Write a program to create an abstract class named shape that contains two integers and an empty method named print Area(). Provide three classes named Rectangle, Triangle and Ellipse such that each one of the classes extends the class shape. Each one of the class contains only the method print Area() that print the area of the given shape

Coding:

```
import java.util.Scanner;

abstract class Shape {
    int h = 0, w = 0;

    abstract void printArea(int h, int w);
}

class Rectangle extends Shape {
    void printArea(int a, int b) {
        h = a;
        w = b;

        float area = h * w;

        System.out.println("The Area of rectangle is " + area);
    }
}

class Triangle extends Shape {
    void printArea(int a, int b) {
        h = a;
        w = b;

        float area = 0.5f * h * w;

        System.out.println("The Area of Triangle is " + area);
    }
}

class Ellipse extends Shape {
    void printArea(int a, int b) {
        h = a;
        w = b;
```

```

        float area = 3.14f * h * h;

        System.out.println("The Area of Ellipse is " + area);
    }
}

class AbClass {
    public static void main(String args[]) {
        Rectangle r = new Rectangle();
        Triangle t = new Triangle();
        Ellipse e = new Ellipse();
        int v1, v2;

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Length and Width of a rectangle");
        v1 = sc.nextInt();
        v2 = sc.nextInt();
        r.printArea(v1, v2);
        System.out.println("*****");
        System.out.print("Enter Base and Height of Traiangle");
        v1 = sc.nextInt();
        v2 = sc.nextInt();
        t.printArea(v1, v2);
        System.out.println("*****");
        System.out.print("Enter axis values of Ellipse");
        v1 = sc.nextInt();
        v2 = sc.nextInt();
        e.printArea(v1, v2);
    }
}

```

Output:

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

G:\Ashwith\Java\LAB PROGRAMS\PART B\B4. Abstract Class>javac AbClass.java

G:\Ashwith\Java\LAB PROGRAMS\PART B\B4. Abstract Class>java AbClass
Enter Length and Width of a rectangle 12 6
The Area of rectangle is 72.0
*****
Enter Base and Height of Traiangle 6 10
The Area of Triangle is 30.0
*****
Enter axis values of Ellipse 20 10
The Area of Ellipse is 1256.0
```

5. Create a package to convert temperature in centigrade into Fahrenheit and one more package to calculate the simple interest. Implement both package in the main() by accepting the required input for each application.

Coding:

```
//Simple Interest Package
```

```
package SimpleIntrest;
```

```
public class SimpleInt {
```

```
    public void SI(int amount, int period, float rate) {
```

```
        float si = (amount * period * rate) / 100;
```

```
        System.out.println("*****");
```

```
        System.out.println("Amount: " + amount);
```

```
        System.out.println("Period: " + period);
```

```
        System.out.println("Rate: " + rate);
```

```
        System.out.println("Simple Interest: " + si);
```

```
        System.out.println("*****");
```

```
    }
```

```
}
```

```
//Temperature Package
```

```
package TempConPack;
```

```
public class CentToFar {
```

```
    public void TempCon(float cels) {
```

```
        float fh;
```

```
        fh = (cels * 1.80f) + 32;
```

```
        System.out.println("Temperature in Celsius: " + cels);
```

```
        System.out.println("Temperature in Fahrenheit: " + fh);
```

```
    }
```

```
}
```

```
//Main Program
```

```
import java.util.Scanner;
```

```
import TempConPack.*;
```

```
import SimpleIntrest.*;
class calculatePack {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        CentToFar cobj = new CentToFar();
        SimpleInt sobj = new SimpleInt();
        int ch = 0;
        float temp = 0.0f, rate = 0.0f;
        int amount, period;
        do {
            System.out.println("1.Temperature Conversion \n2.Simple Intrest \n3.Exit");
            System.out.println("Enter your choice:");
            ch = sc.nextInt();
            switch (ch) {
                case 1:
                    System.out.println("Enter the temperature in Celsius:");
                    temp = sc.nextFloat();
                    cobj.TempCon(temp);
                    break;
                case 2:
                    System.out.println("Enter the amount:");
                    amount = sc.nextInt();
                    System.out.println("Enter a period:");
                    period = sc.nextInt();
                    System.out.println("Enter the rate:");
                    rate = sc.nextFloat();
                    sobj.SI(amount, period, rate);
                    break;
                case 3:
```

```

        System.exit(0);

        break;

    }

    } while (ch <= 3);

}
}

```

Output:

```

Microsoft Windows [Version 10.0.26200.8246]
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```

```

G:\Ashwith\Java\LAB PROGRAMS\PART B\B5. Package\SimpleIntrest>javac SimpleInt.java

```

```

Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

```

```

G:\Ashwith\Java\LAB PROGRAMS\PART B\B5. Package\TempConPack>javac CentToFar.java

```

```

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```

```

G:\Ashwith\Java\LAB PROGRAMS\PART B\B5. Package>javac calculatePack.java

```

```

G:\Ashwith\Java\LAB PROGRAMS\PART B\B5. Package>java calculatePack

```

```

1.Temperature Conversion
2.Simple Intrest
3.Exit

```

```

Enter your choice:

```

```

1

```

```

Enter the temperature in Celsius:

```

```

45

```

```

Temperature in Celsius: 45.0

```

```

Temperature in Fahrenheit: 113.0

```

```

1.Temperature Conversion
2.Simple Intrest
3.Exit

```

```

Enter your choice:

```

```

2

```

```

Enter the amount:

```

```

25000

```

```

Enter a period:

```

```

5

```

```

Enter the rate:

```

```

8

```

```

*****

```

```

Amount: 25000

```

```

Period: 5

```

```

Rate: 8.0

```

```

Simple Intrest: 10000.0

```

```

*****

```

```

1.Temperature Conversion
2.Simple Intrest
3.Exit

```

```

Enter your choice:

```

```

3

```

6. Write a program that implements a multi-threaded program has three threads. First thread generates a random integer every second, and if the value is even, second thread computes the square of the number and prints. If the value is odd the third thread will print the value of cube of the number.

Coding:

```
import java.io.*;
import java.util.*;
class First extends Thread {
    public void run() {
        for (;;) {
            int r;
            Random d = new Random();
            r = d.nextInt(100) + 1;
            try {
                Thread.sleep(1000);
                if (r % 2 == 0) {
                    Thread t2 = new Second(r);
                    t2.start();
                } else {
                    Thread t3 = new Third(r);
                    t3.start();
                }
            } catch (InterruptedException e) {
            }
        }
    }
}
class Second extends Thread {
    int r1;
    Second(int r) {
```



```

        r1 = r;
    }

    public void run() {
        System.out.println("The square of number " + r1 + " is: " + r1 * r1);
    }
}

class Third extends Thread {
    int r1;

    Third(int r) {
        r1 = r;
    }

    public void run() {
        System.out.println("The Cube of the Number " + r1 + " is: " + r1 * r1 * r1);
    }
}

class Mthread {
    public static void main(String[] args) {
        Thread t1 = new First();

        System.out.println("press Ctrl+c to stop.....");

        t1.start();
    }
}

```

Output:

```

Microsoft Windows [Version 10.0.26200.8246]
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G:\Ashwith\Java\LAB PROGRAMS\PART B\B6. Multi Threaded>javac Mthread.java

G:\Ashwith\Java\LAB PROGRAMS\PART B\B6. Multi Threaded>java Mthread
press Ctrl+c to stop.....
The Cube of the Number 7 is: 343
The square of number 24 is: 576
The Cube of the Number 59 is: 205379
The square of number 98 is: 9604
The Cube of the Number 23 is: 12167
The square of number 68 is: 4624
The square of number 4 is: 16

```

7. Write a program that creates a user interface to perform basic integer operations. The user enters two numbers in the textfields, Num1 and Num2. The result of operations must be displayed in the Result textfield when the = button is clicked. If Num1 or Num2 is not an integer, the program should throw NumberFormatException. If Num2 is Zero, the program should throw an ArithmeticException when division operation is applied. Display the exception in a message dialog box.

Coding:

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

class Calculator extends JFrame implements ActionListener {
    JTextField txtNum1, txtNum2, txtOp, txtResult;
    JButton b;
    JLabel lblNum1, lblNum2, lblOp, lblResult;

    Calculator() {
        Container c = getContentPane();
        c.setLayout(new FlowLayout());

        lblNum1 = new JLabel("Number 1 ");
        lblNum2 = new JLabel("Number 2");
        lblOp = new JLabel("Operation");
        lblResult = new JLabel("Result ");
        txtNum1 = new JTextField("", 5);
        txtNum2 = new JTextField("", 5);
        txtOp = new JTextField("", 1);
        txtResult = new JTextField("", 5);
        b = new JButton("=");

        c.add(lblNum1);
        c.add(txtNum1);
        c.add(lblOp);
        c.add(txtOp);
        c.add(lblNum2);
```

```
c.add(txtNum2);  
c.add(b);  
c.add(lblResult);  
c.add(txtResult);  
b.addActionListener(this);  
}  
public void actionPerformed(ActionEvent ae) {  
    if (ae.getActionCommand() == "=") {  
        try {  
            char operator = txtOp.getText().charAt(0);  
            int r = 0, n1 = 0, n2 = 0;  
            n1 = Integer.parseInt(txtNum1.getText());  
            n2 = Integer.parseInt(txtNum2.getText());  
            switch (operator) {  
                case '+':  
                    r = n1 + n2;  
                    break;  
                case '-':  
                    r = n1 - n2;  
                    break;  
                case '*':  
                    r = n1 * n2;  
                    break;  
                case '/':  
                    r = n1 / n2;  
                    break;  
            }  
            txtResult.setText("" + r);  
        } catch (ArithmeticException e1) {
```

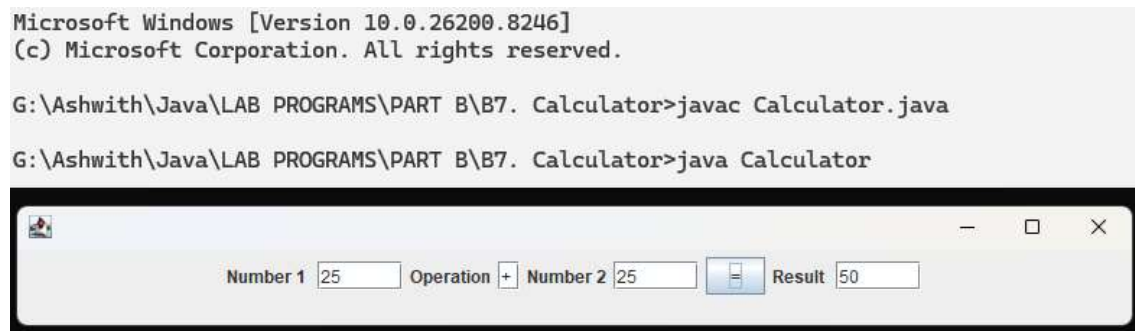
```

        JOptionPane.showMessageDialog(null, "Arithmetic Exception");
        txtResult.setText("");
    } catch (NumberFormatException e2) {
        JOptionPane.showMessageDialog(null, "NumberFormatException");
        txtResult.setText("");
    }
}
}

public static void main(String args[]) {
    Calculator ob = new Calculator();
    ob.setSize(800, 600);
    ob.setVisible(true);
    ob.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

```

Output:



8. Using the swing components, design the frame for shopping a book that accepts book code, book name, and Price. Calculate the discount on code as follows.

Code	Discount rate
-------------	----------------------

101	15%
------------	------------

102	20%
------------	------------

103	25%
------------	------------

Any other	5%
------------------	-----------

Find the discount amount and Net bill amount. Display the bill.

Coding:

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

public class Shoppingbook implements ActionListener {

    JLabel lcode;
    JLabel lname;
    JLabel lprice;
    JLabel ldiscount;
    JLabel lnetbill;
    JTextField tcode;
    JTextField tname;
    JTextField tprice;
    JTextField tdiscount;
    JTextField tnetbill;
    JButton bill;

    Shoppingbook() {

        JFrame jfrm = new JFrame("Book Shopping bill");
        jfrm.setLayout(null);
        jfrm.setSize(300, 300);
        jfrm.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
```

```
lcode = new JLabel("Book Code");
tcode = new JTextField();
lname = new JLabel("Book Name");
tname = new JTextField();
lprice = new JLabel("Book Price");
tprice = new JTextField();
ldiscount = new JLabel("Discount");
tdiscount = new JTextField();
lnetbill = new JLabel("Final Price");
tnetbill = new JTextField();
bill = new JButton("Bill");
lcode.setBounds(20, 20, 80, 10);
tcode.setBounds(100, 18, 100, 20);
lname.setBounds(20, 50, 80, 10);
tname.setBounds(100, 48, 100, 20);
lprice.setBounds(20, 80, 80, 10);
tprice.setBounds(100, 78, 100, 20);
ldiscount.setBounds(20, 110, 80, 10);
tdiscount.setBounds(100, 108, 100, 20);
lnetbill.setBounds(20, 140, 80, 10);
tnetbill.setBounds(100, 138, 100, 20);
bill.setBounds(100, 180, 60, 30);
jfrm.add(lcode);
jfrm.add(tcode);
jfrm.add(lname);
jfrm.add(tname);
jfrm.add(lprice);
jfrm.add(tprice);
jfrm.add(ldiscount);
```

```

jfrm.add(tdiscount);
jfrm.add(lnetbill);
jfrm.add(tnetbill);
bill.addActionListener(this);
jfrm.add(bill);
jfrm.setVisible(true);
tdiscount.setEditable(false);
}
public void actionPerformed(ActionEvent ae) {
    double discount, price, netbill;
    String discountamt, netbillamt;
    try {
        int codeval = Integer.parseInt(tcode.getText());
        if (codeval == 101)
            discount = 15;
        else if (codeval == 102)
            discount = 20;
        else if (codeval == 103)
            discount = 25;
        else
            discount = 5;
        discountamt = Double.toString(discount);
        if (!(tcode.getText().equals("")) && !(tname.getText().equals("")) &&
            !(tprice.getText().equals("")))
            tdiscount.setText(discount + "%");
        price = Integer.parseInt(tprice.getText());
        netbill = price - (price * discount / 100);
        netbillamt = Double.toString(netbill);
        tnetbill.setText(netbillamt);
    }
}

```

```

    } catch (NumberFormatException e) {
        tnetbill.setText("Invalid number");
    }
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(new Runnable() {
        public void run() {
            new Shoppingbook();
        }
    });
}
}

```

Output:

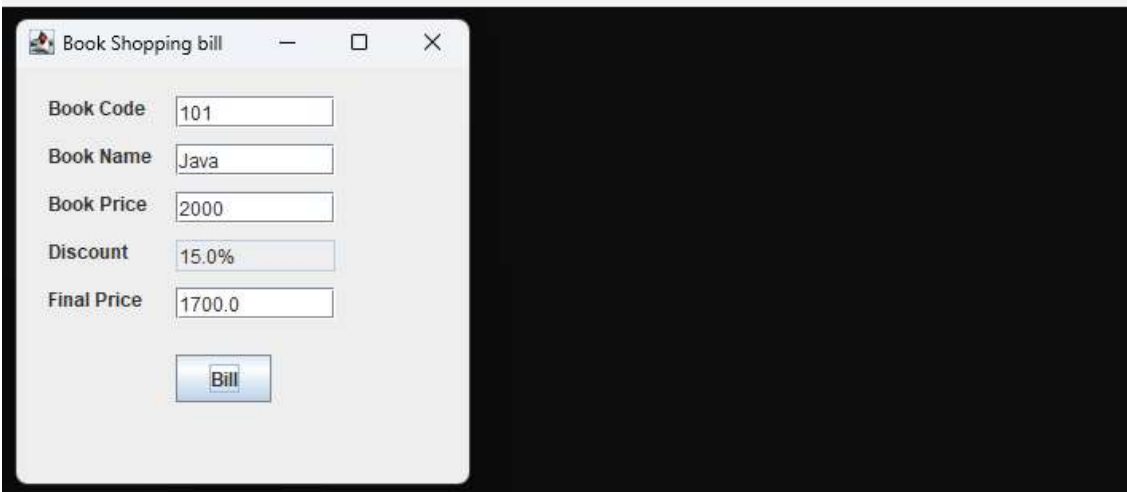
```

Microsoft Windows [Version 10.0.26200.8246]
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G:\Ashwith\Java\LAB PROGRAMS\PART B\B8. Shopping Book>javac Shoppingbook.java

G:\Ashwith\Java\LAB PROGRAMS\PART B\B8. Shopping Book>java Shoppingbook

```



The screenshot shows a Java Swing window titled "Book Shopping bill". It features a light gray background and a white border. Inside the window, there are five text input fields arranged vertically, each with a label to its left. The labels and their corresponding values are: "Book Code" with "101", "Book Name" with "Java", "Book Price" with "2000", "Discount" with "15.0%", and "Final Price" with "1700.0". Below these fields is a single button labeled "Bill". The window has standard Windows window controls (minimize, maximize, close) in the top right corner.